



ENTOFARM

IMPLEMENTING CIRCULAR ECONOMY PRACTICES IN DAILY LIFE

<https://www.ento.farm/>



MODULE OVERVIEW

The linear economic model (take-make-dispose) has driven progress but also leads to major environmental problems such as resource depletion and waste. Global resource extraction has tripled since 1970 and could increase by 60% by 2060 (UN).

The circular economy offers an alternative by promoting resource reuse and turning waste into a resource. This module introduces these ideas through everyday examples, helping students understand sustainability and develop responsible, creative skills for the green transition.

TARGET GROUP

Age Range: 12–18 years

Subject Links: STEM, Environmental Science, Geography, Civic Education, Economics

LEARNING OBJECTIVES

By the end of this module, students will:

Knowledge: Understand what the circular economy is, how it differs from the linear model, and the key principles of reduce, reuse, repair, repurpose, and recycle.

Skills: Analyze everyday examples of resource use and waste and collaboratively propose simple circular solutions.

Attitudes: Develop a sustainability mindset, responsible consumption habits, and appreciation for cooperation, innovation, and environmental responsibility.



SCAN ME

Who are we?

About the Project

This module has been developed within the framework of an **Erasmus+ proposal** focusing on **sustainable agriculture, circular economy, and innovative learning** approaches in STEM education.

The project brings together educational institutions, universities, and organizations from **Greece, Bulgaria and North Macedonia** to develop engaging educational materials for secondary school students. Its main objective is to help young people better understand how sustainable practices can address environmental challenges while **encouraging interest in STEM education and career**.

A central element of the project is the EntoFARM educational concept, which introduces students to alternative and sustainable food production systems such as insect farming and other circular resource management practices. Through these examples, students explore how biological processes, technological innovation, and responsible resource use can contribute to more sustainable food systems.

The educational materials produced in the project aim to support teachers in introducing sustainability and circular economy concepts through interactive and interdisciplinary learning activities.



Project Consortium:

- Fraud Line – Greece (Project Coordinator)
- University of Thessaly – Greece
- University of Patras – Greece
- 1st EPAL Volou – Greece
- Professional Agricultural School “Buzema” – Bulgaria
- Professional Forum for Education – Bulgaria
- Association for Internationalization of Education and Science – North Macedonia

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FROM LINEAR TO CIRCULAR THINKING

Our traditional linear economy follows a simple path: **extract** → **produce** → **consume** → **dispose**.

This model assumes that natural resources are unlimited and waste can be endlessly absorbed by the planet. However, this is no longer true. A part of this linear model is planned obsolescence – when a product has been designed to have a limited lifespan to encourage consumers to buy it again, which is contradictory to sustainable development principles.

In a circular economy, materials are kept in use for as long as possible. Instead of being thrown away, products and components are designed to be reused, repaired, or recycled. The goal is not just to reduce waste, but to eliminate the very concept of waste by thinking in closed loops.

Linear Economy	Circular Economy
Take → Make → Dispose	Design → Use → Reuse/Repair → Recycle
Finite resources	Regenerative cycles
Waste as inevitable	Waste as a design flaw
Focus on production	Focus on systems and longevity

Example:

In a linear system, a smartphone is produced, used for two years, and discarded.

In a circular system, it is designed for easy repair, components are upgraded, and materials are recovered for new phones.



CORE PRINCIPLES OF THE CIRCULAR ECONOMY



1. Reduce

Reducing means using fewer resources in the first place. It focuses on preventing waste before it happens - buying less, choosing durable products, and minimizing energy use.

Example: Bringing a reusable water bottle instead of buying single-use plastic ones.

In the classroom: Students can calculate how much plastic they would save if each person avoided one disposable bottle per day for a school year.



2. Reuse

Reusing extends the lifespan of products and materials by finding new purposes for them without significant processing.

Example: Passing down clothes to younger siblings, refilling jars, or using one-sided printed paper for notes.

Activity idea: "Reusability Relay" - groups brainstorm how to reuse 10 common items (e.g., jars, boxes, plastic bags, clothes).



3. Repair

Repairing encourages fixing broken items instead of discarding them. It also builds technical skills and appreciation for craftsmanship.

Example: Repair cafés - community spaces where volunteers fix broken electronics, bikes, or clothes - embody the repair principle.

Discussion: Why do people often replace instead of repair products? (answers may include convenience, design, cost, or marketing).

CORE PRINCIPLES OF THE CIRCULAR ECONOMY



4. Repurpose / Upcycle

Repurposing means transforming used materials into products of equal or greater value. Unlike recycling, which often downgrades material quality, upcycling gives items new life through creativity.

Example: Making planters from old tires, jewelry from electronic parts, or furniture from pallets.

Class activity: Ask students to sketch or prototype an “upcycled” product they could make at home.



5. Recycle

Recycling converts waste into new materials, closing the loop. However, it should be the last step – after reduction, reuse, and repair – since recycling still requires energy.

Example: Aluminum cans can be infinitely recycled with minimal quality loss, unlike plastics that degrade after a few cycles.



6. Regenerate

The most advanced principle of circularity is regeneration – restoring and enriching natural systems. It’s not enough to do “less harm”; the goal is to do “more good.”

Example: Composting food waste returns nutrients to the soil. Planting trees or maintaining green roofs helps regenerate ecosystems.

Reflection question: What would a regenerative school look like?

REAL WORLD APPLICATIONS

Understanding circular economy principles becomes more meaningful when students see how these ideas are applied in the world around them. Below are examples at three scales - individual, community, and business/industry - with discussion prompts teachers can use.

How it looks in practice:



Individual Level

At home or school, everyone can apply circular principles through daily habits.

Examples:

- **Refuse:** Decline unnecessary packaging or freebies that quickly become waste.
- **Reduce:** Buy less and choose high-quality items that last.
- **Reuse:** Carry reusable shopping bags, bottles, and lunch boxes.
- **Repair:** Sew torn clothes or fix a loose handle before buying new.
- **Recycle:** Sort waste correctly and learn which materials are recyclable locally.
- **Regenerate:** Compost food scraps or plant herbs in upcycled containers.

Class discussion prompt:

“What is one small change you could make this week that would reduce waste in your home?” Encourage students to make a short list and reflect at the end of the week on whether they managed to apply it.



Community Level:

Communities and local governments can also implement circular practices through shared initiatives.

Examples:

- Repair cafés and swap events where residents fix or exchange goods.
- Community composting projects that turn organic waste into fertilizer.
- Tool libraries that allow people to borrow equipment rather than buy new.
- School sustainability clubs that organize “Zero Waste Days” or recycling campaigns.

Project idea:

Students can map all local circular initiatives in their city or village and present them on a community “Circular Map.” This helps them see how circular practices exist outside the classroom.

REAL WORLD APPLICATIONS



Business & Industry Level

At the business and industry level, circular economy means redesigning how products, services, and supply chains work. Companies move away from selling disposable products and instead focus on durability, reuse, waste valorization, and service-based models.

Business models to include:

Product-as-a-Service: Companies keep ownership; users access the product

- **Example:** leasing electronics, shared tools, bike-sharing

Industrial Symbiosis: Waste from one company becomes input for another

- **Example:** food waste → animal feed → fertilizer

Circular Bioeconomy: Biological resources regenerated instead of extracted

- **Example:** composting, bioenergy, insect farming

MISCONCEPTIONS TO ADDRESS

Misconception		Clarification
"Circular economy is just another word for recycling."	➔	Recycling is only one part; the circular economy starts with design and prevention.
"It's too expensive to implement."	➔	Many circular actions save money over time (e.g., reuse, repair, bulk buying).
"Only governments and big companies can make a difference."	➔	Individual and community actions create the demand that drives systemic change.
"Circular products are always more sustainable."	➔	Not necessarily - students must think critically about materials, transport, and life cycles.

CASE STUDY:

FOODOBOX (<https://foodobox.com/>)

Circular strategy/ Background:

Food waste prevention and redistribution.

In the food sector, large quantities of edible food are wasted every day due to overproduction, unsold stock, or approaching expiration dates. This waste represents lost resources such as water, energy, and labor.



What the company does and what circular economy principles were applied?

Foodobox is a Bulgarian digital platform that connects restaurants, bakeries, and food retailers with consumers. Businesses offer surplus, perfectly edible food at a reduced price instead of throwing it away. Customers order food boxes through the platform and collect them at a set time.

Circular economy principles applied:

Reduce: Prevents edible food from becoming waste

Reuse: Redirects surplus food to consumers

Systems thinking: Uses digital tools to connect supply and demand



Sector:

Food services and retail

Why this example matters?

This case study demonstrates how local businesses can apply circular economy principles effectively. It shows that waste prevention is one of the most impactful circular actions and that circular solutions do not require complex technology—only smart organization and cooperation.



Reflection questions:

- Why is preventing food waste more effective than recycling it?
- How does Foodobox benefit both businesses and consumers?
- Could a similar model work in school cafeterias or local communities?

CASE STUDY: PATAGONIA

Circular strategy/ Background:

Durability, repair, and reuse

The fashion industry is one of the most resource-intensive industries and generates large amounts of textile waste. Fast fashion models encourage frequent consumption and disposal of clothing.



What the company does and what circular economy principles were applied?

Patagonia designs clothing intended for long-term use. The company actively promotes repairing and reusing garments instead of replacing them. It offers repair services, repair guides for customers, and a second-hand resale platform for used Patagonia products.

Circular economy principles applied:

Reduce: Encourages mindful consumption

Repair: Extends clothing lifespan

Reuse: Keeps textiles in use through resale

Design for longevity: Focus on quality and durability



Sector:

Clothing and textiles

Why this example matters?

This case study challenges the idea that business success depends on selling more products. It shows that promoting repair and reuse can be part of a viable business model while reducing environmental impact.



Reflection questions:

- Why has clothing repair become less common?
- How does this approach differ from fast fashion?
- Would easier access to repair services change consumer behavior?

CASE STUDY: PROTIX

Circular strategy/ Background:

Waste-to-resource and regenerative bioeconomy

Food systems produce large amounts of organic waste, while conventional animal feed production relies heavily on soy and fishmeal, contributing to deforestation and overfishing.



What the company does and what circular economy principles were applied?

Protix uses organic food waste as feed for insects, mainly black soldier fly larvae. The insects are processed into protein-rich animal feed and insect oil. The remaining by-product (frass) is used as a natural fertilizer, returning nutrients to the soil.

Circular economy principles applied:

Repurpose: Transforms food waste into valuable products

Recycle (biological loop): Nutrients return to the food system

Regenerate: Supports soil health through organic fertilizer



Sector:

Agriculture and food systems

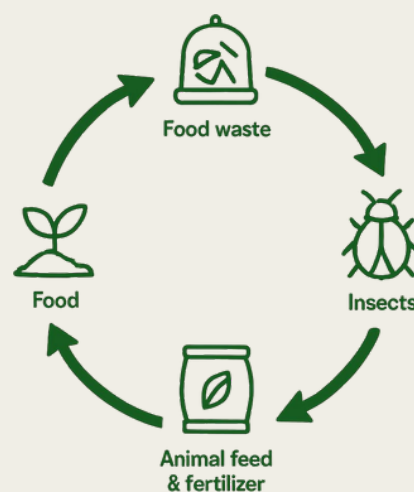
Why this example matters?

This case study illustrates a closed-loop biological system and shows how circular economy principles can be applied at an industrial scale. It directly connects circular economy thinking with sustainable agriculture and food security.



Reflection questions:

- What resources are saved through insect farming compared to traditional feed production?
- How does this model reduce pressure on ecosystems?
- Could insect farming be applied at a local or school level?



PROPOSED SCHOOL ACTIVITY

“Closing the Loop: The Classroom Waste Audit”

Background

Purpose: To explore how students can apply circular economy principles through direct investigation of their own classroom waste, identifying how everyday habits contribute to resource loss - and how small actions can close the loop.

Relevance: This case study connects the circular economy with tangible student behavior. It shows that systemic change starts from awareness and evidence - not abstract ideas.

It also demonstrates citizen science in action: young people collecting and interpreting real data to inform their community’s environmental choices.

Setting

A secondary school classroom where a science and civic education teacher decides to conduct a “mini waste audit” as part of the circular economy lesson.

The goal: to find out what types of waste the class produces in one day - and brainstorm ways to reduce or reuse them.

Characters

- Ms. Georgieva – Environmental Science teacher, passionate about practical sustainability.
- Class 9B – A lively group of 15–16-year-olds with mixed levels of environmental awareness.
- Three Student Teams:
 - Collectors – gather the day’s waste.
 - Sorters – categorize it into material types.
 - Analysts – record and interpret data.

Scenario Summary

During the week’s environmental module, Ms. Georgieva challenges her students to analyze their classroom waste instead of just talking about recycling. “Let’s see what our trash can tell us,” she says. The students agree to collect all their waste - paper scraps, snack wrappers, bottles, pens - for one school day. The next day, they bring gloves, a scale, and data sheets to conduct their own Classroom Waste Audit.

PROPOSED SCHOOL ACTIVITY

School Activity: “Closing the Loop: The Classroom Waste Audit”

Detailed Scenario

The students begin by emptying their classroom trash bin onto a covered table (under teacher supervision).

They quickly discover that most of what they throw away isn't hazardous or messy - it's paper, plastic bottles, and candy wrappers.

After sorting the waste into categories (paper, plastic, metal, organic, mixed), they weigh each pile and note the results.

Their findings surprise them:

- **40%** of the waste is paper, mostly one-sided printed worksheets and doodled notes.
- **30%** is plastic packaging, mainly snack wrappers and bottles.
- **15%** is metal cans from drinks.
- **10%** is organic waste (food).
- **5%** is mixed or non-recyclable items, such as pen cartridges or tissues.

The analysts calculate that, if every class in the school generated similar amounts, the school produces over 30 kilograms of classroom waste per week - much of which could be avoided or reused.

When Ms. Georgieva asks, “So, what does this tell us about our habits?”, the discussion heats up.

Students realize they often buy single-use packaging such as plastic bottles and they don't recycle the waste.

One student, Emil, suggests:

“We could encourage everyone to bring a reusable water bottle with them to school, as well as to bring home-made food for the lunch breaks in a reusable recipient. Maybe we can put posters on walls or go to each class to speak to the students about why this is important”

Another, Daria, adds:

“Let's have a ‘Plastic-Free Friday’ challenge for the whole school – no plastic packaging at all. And let's speak with the headmaster to put separate bins for paper, plastic, metal and glass in the school corridors”

They agree to present their results and recommendations to the student council and the headmaster.

Within a month, their ideas are adopted - the school develops a campaign to encourage the use of reusable water bottles and food containers, and sets up trash bins in the corridors that support students to separate their trash which is then recycled.

The small classroom initiative becomes a whole-school circular action.

PROPOSED SCHOOL ACTIVITY

School Activity: “Closing the Loop: The Classroom Waste Audit”

Discussion Questions

1. What were the main types of waste found in the classroom?
2. Which circular economy principles (reduce, reuse, recycle, regenerate) can you identify in the students’ actions?
3. What behavioral or design changes could reduce these waste types?
4. What challenges might the students face when trying to implement their recommendations?
5. How could this waste audit model be scaled to the whole school or community?
6. What role can data play in changing environmental behavior?

Follow-up idea: You can organize a similar classroom waste audit in your school, in order to compare the results of the analysis with the case study ones. Then think about ideas for reducing this waste, such as composting organic waste, separating and recycling, encouraging students to use reusable water bottles, etc.

ASSESSMENT & EVALUATION

Quiz: Circular Economy Knowledge Check

Multiple Choice

1. What is a core aim of the circular economy?

- a) To increase annual production of goods
- b) To keep materials and products in use for as long as possible
- c) To encourage people to buy more
- d) To exclusively focus on renewable energy

2. According to circular economy strategies, what should be done with products at the end of their life?

- a) Throw them away in landfill
- b) Keep them in your house
- c) Recycle, remanufacture, or redesign them
- d) Burn them

3. Which of the following is an example of a circular economy solution?

- a) Designing products for shorter lifespans
- b) Increasing single-use packaging
- c) Supporting local farming and reducing excess consumption
- d) Encouraging more travel

4. What principle can help reduce food waste and improve sustainability at home?

- a) Buy more processed foods
- b) Cook creatively
- c) Ignore expiry dates
- d) Use only imported foods

5. Which of these best describes the linear economy model that the circular economy seeks to change?

- a) Take → Make → Waste
- b) Reuse → Repair → Recycle
- c) Design → Share → Remanufacture
- d) Produce → Repair → Regenerate

6. How can consumers help advance the circular economy?

- a) Buy as many new products as possible
- b) Choose products designed to be repaired or reused
- c) Throw all materials in general waste
- d) Only use products once

7. Which strategy is not considered part of building a circular economy according to the solutions list?

- a) Reducing excessive consumption
- b) Supporting local production
- c) Eating healthier foods with lower resource use
- d) Increasing planned obsolescence

ASSESSMENT & EVALUATION

Quiz: Circular Economy Knowledge Check

Multiple Choice

8. What is one reason the circular economy matters for the environment?

- a) It increases resource extraction
- b) It eliminates all economic activities
- c) It cuts waste and helps reduce greenhouse gas emissions
- d) It reduces job opportunities

9. Which action best reflects circular economy thinking in daily life?

- a) Replacing products as soon as a newer version appears
- b) Choosing disposable items for convenience
- c) Repairing or reusing items instead of buying new ones
- d) Throwing recyclable materials in mixed waste

10. According to circular economy approaches, which change could significantly reduce waste at a system level?

- a) Designing products that are difficult to repair
- b) Encouraging consumers to buy more frequently
- c) Shifting from ownership to sharing or service-based models
- d) Increasing single-use packaging for hygiene reasons

ADDITIONAL RESOURCES

Links:

- Ellen MacArthur Foundation: [Circular Economy Schools Resources](#)
- Ellen MacArthur Foundation short video on circular economy:
<https://www.youtube.com/watch?v=zCRKvDyyHmI>
- TED-Ed Video: How the Circular Economy Works (3:15)
- Eco-Future Manual:
<https://circulareconomy.europa.eu/platform/sites/default/files/2024-09/ECO-FUTURE-Manual.pdf>
- Eco-Future Toolkit:
<https://circulareconomy.europa.eu/platform/sites/default/files/2024-09/ECOFUTURE-TOOLKIT.pdf>
- Circular Economy: definition, importance and benefits (European Parliament):
<https://www.europarl.europa.eu/topics/en/article/20151201STO05603/circular-economy-definition-importance-and-benefits>

TEACHER REFLECTION AND ADAPTATION NOTES

This module has been designed as a flexible educational resource that can be adapted to different classroom contexts, student backgrounds, and educational levels.

Teachers are encouraged to reflect on how the activities and content can be adjusted to best support the needs of their students.

Reflection Questions for Teachers

Before or after implementing the module, teachers may consider the following questions:

- Did the activities engage students with different interests and learning styles?
- Were all students encouraged to participate actively in discussions and group work?
- Did the module help students understand the connection between science, technology, and sustainability?
- Were examples and explanations accessible to students with different levels of prior knowledge?

Accessibility Considerations

To ensure that the module is accessible to all learners, teachers may consider:

- simplifying technical terminology when necessary
- providing visual aids, diagrams, or short videos
- allowing students to express their understanding through different formats (presentations, posters, discussions, or written reflections)
- offering additional guidance or scaffolding for students who may need more support

Supporting Girls' Engagement in STEM

Research shows that girls are more likely to engage with STEM when learning environments are collaborative and when scientific topics are linked to real-world challenges and social impact.

Teachers can therefore:

- encourage mixed-gender teamwork and equal participation
- highlight female role models in sustainability science and agriculture
- frame STEM challenges as problem-solving activities that benefit society and the environment
- ensure that classroom discussions give equal voice to all students

TEACHER REFLECTION AND ADAPTATION NOTES

Adapting the Module

Teachers are free to adapt the module by:

- adjusting the depth of scientific explanations
- modifying activities to fit available time or resources
- connecting the topic to local environmental or agricultural examples
- integrating the module with subjects such as biology, geography, environmental science, or technology

The goal is to ensure that the module remains relevant, inclusive, and meaningful for students in different educational settings.

Do you have any questions?

Please let us know! We are happy to help.

Contact us at:



<https://www.ento.farm/>



entofarmproject@gmail.com



+30211-1155525



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